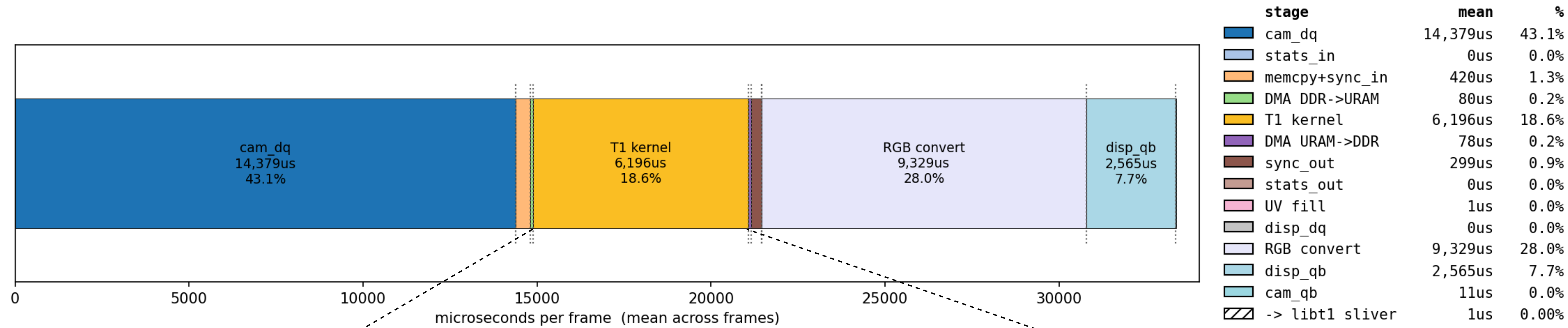
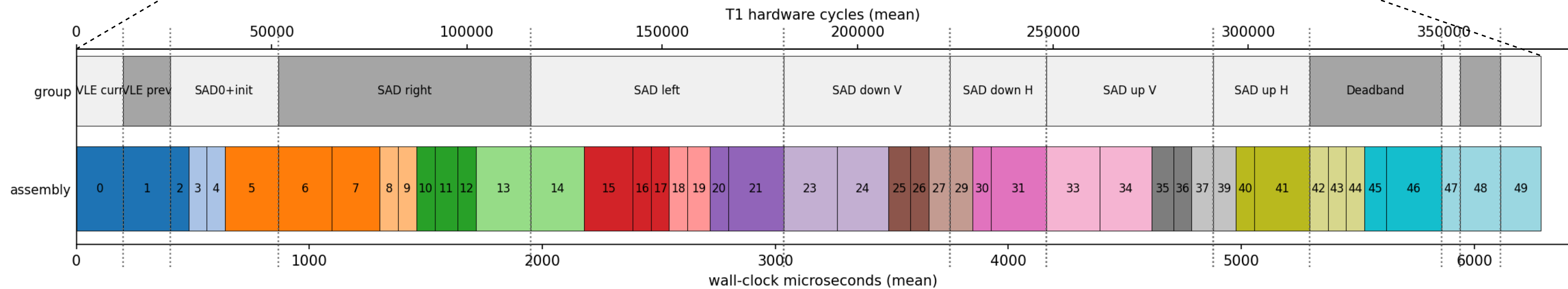


VisionSoC pipeline per-frame breakdown N=10000 frames (skipping frame 0 warm-up)
mean frame = 33,358us ~30.0 fps T1 kernel = 6,196us wall = 6,196us T1@60MHz + 1us libt1/Linux



Optical-flow kernel (5-direction block matching): per-instruction breakdown 46 issued ops + 4 dispatcher-only CSR flips
total T1 cycles = 374.8k total wall = 6,285us libt1 sliver = 39us (@60MHz)



stage	assembly	group	cycles	wall
	0: vle8.v	VLE curr	12.1k	202us
	1: vle8.v	VLE prev	12.1k	202us
	2: vsub.vv	SAD0+init	4.7k	79us
	3: vrsb.vi	SAD0+init	4.6k	77us
	4: vmax.vv	SAD0+init	4.7k	79us
	5: vmv.v.i	SAD0+init	13.6k	228us
	6: vmv.v.i	SAD right	13.6k	228us
	7: vslideup.vi	SAD right	12.4k	207us
	8: vsub.vv	SAD right	4.7k	79us
	9: vrsb.vi	SAD right	4.6k	77us
	10: vmax.vv	SAD right	4.7k	79us
	11: vmsltu.vv	SAD right	5.8k	97us
	12: vminu.vv	SAD right	4.7k	79us
	13: vmerge.vim	SAD right	14.2k	237us

stage	assembly	group	cycles	wall
	14: vmv.v.i	SAD left	13.6k	228us
	15: vslidedown.vi	SAD left	12.5k	209us
	16: vsub.vv	SAD left	4.7k	79us
	17: vrsb.vi	SAD left	4.6k	77us
	18: vmax.vv	SAD left	4.7k	79us
	19: vmsltu.vv	SAD left	5.7k	97us
	20: vminu.vv	SAD left	4.7k	79us
	21: vmerge.vim	SAD left	14.2k	237us
	22: csrwi(vmode=1)	CSR flip	0	0us
	23: vmv.v.i	SAD down V	13.6k	228us
	24: vslideup.vi	SAD down V	13.3k	222us
	25: vsub.vv	SAD down V	5.6k	94us
	26: vrsb.vi	SAD down V	4.6k	77us
	27: vmax.vv	SAD down V	5.6k	94us
	28: csrwi(vmode=0)	CSR flip	0	0us
	29: vmsltu.vv	SAD down H	5.7k	97us
	30: vminu.vv	SAD down H	4.7k	79us
	31: vmerge.vim	SAD down H	14.2k	237us

stage	assembly	group	cycles	wall
	32: csrwi(vmode=1)	CSR flip	0	0us
	33: vmv.v.i	SAD up V	13.7k	229us
	34: vslidedown.vi	SAD up V	13.4k	224us
	35: vsub.vv	SAD up V	5.6k	94us
	36: vrsb.vi	SAD up V	4.6k	77us
	37: vmax.vv	SAD up V	5.6k	94us
	38: csrwi(vmode=0)	CSR flip	0	0us
	39: vmsltu.vv	SAD up H	5.7k	96us
	40: vminu.vv	SAD up H	4.7k	79us
	41: vmerge.vim	SAD up H	14.2k	237us
	42: vsub.vv	Deadband	4.7k	79us
	43: vrsb.vi	Deadband	4.6k	77us
	44: vmax.vv	Deadband	4.7k	79us
	45: vmsleu.vi	Deadband	5.6k	94us
	46: vmerge.vim	Deadband	14.2k	237us
	47: vmul.vx	Scale x50	4.8k	81us
	48: vse8.v	VSE motion	10.3k	173us
	49: vse8.v	VSE prev<-curr	10.3k	172us